

Random Vectors

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Objectives of the lecture

Define four modes of convergence for random vectors.

Characterise convergence in distribution without cumulative distribution functions. The portmanteau theorem replaces them by test functions and by sets, which is what we use in every later proof.

Turn it into an analytic condition. Characteristic functions convert convergence of laws into pointwise convergence of functions on $\mathbb{R}^d$.

Assemble a calculus. Slutsky, continuous mapping, and the symbols $o_{\mathbb{P}}$ and $O_{\mathbb{P}}$.

Notation for this lecture

The space. All random vectors live on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and take values in $\mathbb{R}^d$ with the Euclidean inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. We denote measures by capital blackboard letters ($\mathbb{P}$ on Ω , $\mathbb{P}_X$ for the law of X on $\mathbb{R}^d$, $\mathbb{P}^{\otimes n}$ for a product) and densities by lowercase letters (p, p_θ) .

Subscripts. Every expectation, variance and supremum carries what it is taken over: $\mathbb{E}_X$, Var_X , $\mathbb{E}_{X_n}$, and $\mathbb{E}_S$ for a sample $S = (X_1, \dots, X_n) \sim \mathbb{P}_X^{\otimes n}$.

Arrows. $X_n \xrightarrow{d} X$, $X_n \xrightarrow{\mathbb{P}} X$, $X_n \xrightarrow{a.s.} X$, $X_n \xrightarrow{L^p} X$, all as $n \rightarrow \infty$.

Sets. For $A \subset \mathbb{R}^d$, $\bar{A}$ is the closure, $\mathring{A}$ the interior, $\partial A = \bar{A} \setminus \mathring{A}$ the boundary and A^c the complement. We also set $B(c, r) = \{x : \|x - c\| < r\}$.

Part I. Four modes of convergence

The cumulative distribution function

Definition. For a random vector $X = (X_1, \dots, X_d)$ of $\mathbb{R}^d$, its cumulative distribution function is

$$F_X : \mathbb{R}^d \rightarrow [0, 1], \quad (x_1, \dots, x_d) \mapsto \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d).$$

Proposition. F_X determines $\mathbb{P}_X$.

Proof. The family $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}^d\}$, where $(-\infty, x] = \prod_{j \leq d} (-\infty, x_j]$, is stable under finite intersections and generates $\mathcal{B}(\mathbb{R}^d)$. By the Dynkin lemma, two laws agreeing on a generating π -system agree on the σ -field it generates, which concludes the proof.

Four modes of convergence

Definition. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random vectors of $\mathbb{R}^d$ and X a random vector of $\mathbb{R}^d$. We say that X_n converges to X

in distribution, written $X_n \xrightarrow{d} X$, if $F_{X_n}(x) \rightarrow F_X(x)$ for every x at which F_X is continuous.

in probability, written $X_n \xrightarrow{\mathbb{P}} X$, if $\mathbb{P}(\|X_n - X\| \geq \epsilon) \rightarrow 0$ for every $\epsilon > 0$.

almost surely, written $X_n \xrightarrow{a.s.} X$, if $\mathbb{P}(\|X_n - X\| \rightarrow 0) = 1$.

in L^p , $p \geq 1$, written $X_n \xrightarrow{L^p} X$, if $\mathbb{E}_{X_n, X}[\|X_n - X\|^p] \rightarrow 0$.

Requirements of each definition

Only convergence in distribution is a statement about laws. Indeed, it involves $\mathbb{P}_{X_n}$ and $\mathbb{P}_X$ and nothing else, so that the X_n need not be defined on a common probability space, or on any common space with X . In contrast, the other three use $X_n - X$, and hence require all of them on one $(\Omega, \mathcal{A}, \mathbb{P})$.

Consequence. The limit in distribution can be a law $\mathcal{L}$ on $\mathbb{R}^d$ rather than a random vector, and we write $X_n \xrightarrow{d} \mathcal{L}$. Every statement below holds with that reading, with $\mathbb{P}(X \in A)$ replaced by $\mathcal{L}(A)$.

A limit in distribution is unique as a law, but not as a random vector. Indeed, if $X_n \xrightarrow{d} X$ and $X' \stackrel{d}{=} X$, then $X_n \xrightarrow{d} X'$, whatever the dependence between X' and the X_n .

Restriction to continuity points

Let us take $d = 1$, $X_n = 1/n$ deterministic and $X = 0$. Then $X_n \rightarrow X$ in every reasonable sense, and yet

$$F_{X_n}(0) = \mathbb{P}(1/n \leq 0) = 0 \not\rightarrow 1 = F_X(0).$$

Note that 0 is the unique discontinuity point of F_X . Requiring convergence there would make even a deterministic sequence converging to a point fail to converge in distribution, hence the definition excludes such points. However, at every $x \neq 0$, we do have $F_{X_n}(x) \rightarrow F_X(x)$.

Remark. The set of discontinuity points of F_X is small. Indeed, for each coordinate j , at most countably many t satisfy $\mathbb{P}(X_j = t) > 0$, and F_X is continuous at every x whose coordinates avoid those countable sets. As a result, the definition still constrains F_{X_n} on a dense set.

Countability of the atoms. Let us fix j and $m \geq 1$, and let $A_m = \{t : \mathbb{P}(X_j = t) > 1/m\}$. Distinct $t \in A_m$ correspond to disjoint events $\{X_j = t\}$, whose probabilities sum to at most 1, so that A_m has fewer than m elements. Hence, the atoms of X_j form $\bigcup_{m \geq 1} A_m$, a countable union of finite sets.

Part II. The portmanteau theorem

The portmanteau theorem

Theorem. Let (X_n) and X be random vectors of $\mathbb{R}^d$. The following are equivalent.

- (i) $X_n \xrightarrow{d} X$.
- (ii) $\mathbb{E}_{X_n}[f(X_n)] \rightarrow \mathbb{E}_X[f(X)]$ for every bounded continuous $f : \mathbb{R}^d \rightarrow \mathbb{R}$.
- (iii) $\mathbb{E}_{X_n}[f(X_n)] \rightarrow \mathbb{E}_X[f(X)]$ for every bounded Lipschitz f .
- (iv) $\liminf_n \mathbb{E}_{X_n}[f(X_n)] \geq \mathbb{E}_X[f(X)]$ for every continuous $f \geq 0$.
- (v) $\liminf_n \mathbb{P}(X_n \in O) \geq \mathbb{P}(X \in O)$ for every open O .
- (vi) $\limsup_n \mathbb{P}(X_n \in F) \leq \mathbb{P}(X \in F)$ for every closed F .
- (vii) $\mathbb{P}(X_n \in B) \rightarrow \mathbb{P}(X \in B)$ for every Borel B with $\mathbb{P}(X \in \partial B) = 0$.

Practical use of the conditions

(i) is never used in practice, since cumulative distribution functions are unusable in $d > 1$ and generally impractical.

(ii) and (iii) to prove that something converges, because expectations of test functions are what characteristic functions, Slutsky and continuous mapping all produce.

(vi) to bound a probability from above by the limit, which is how we prove tightness and continuous mapping.

(vii) to read off limit probabilities, which is how we justify a confidence interval in Part VI.

Interest of (iii). The class of bounded Lipschitz functions is much smaller than the class of bounded continuous ones, so that (iii) is the easiest sufficient condition to verify.

Proof, from (i) to convergence on rectangles

We first assume F_X continuous everywhere. For $a \leq b$ in $\mathbb{R}^d$ and $(a, b] = \prod_{j \leq d} (a_j, b_j]$, inclusion and exclusion over the 2^d corners yields

$$\mathbb{P}(X \in (a, b]) = \sum_{\epsilon \in \{0,1\}^d} (-1)^{|\epsilon|} F_X(c_\epsilon), \quad (c_\epsilon)_j = \epsilon_j a_j + (1 - \epsilon_j) b_j,$$

and the same identity holds for each X_n . Since every corner is a continuity point, we obtain $\mathbb{P}(X_n \in (a, b]) \rightarrow \mathbb{P}(X \in (a, b])$.

For a rectangle with a closed left face, we have

$$\mathbb{P}(a < X_n \leq b) \leq \mathbb{P}(a \leq X_n \leq b) \leq \mathbb{P}(a - 2^{-m} \mathbf{1} < X_n \leq b),$$

and letting $n \rightarrow \infty$ on both ends, then $m \rightarrow \infty$, we close the sandwich by continuity of F_X . The other faces are treated identically.

Proof, from rectangles to test functions

Let us fix $\epsilon > 0$ and a compact rectangle I_ϵ with $\mathbb{P}(X \notin I_\epsilon) \leq \epsilon$, which exists since the events $A_M = \{X \in [-M, M]^d\}$ increase along $M \in \mathbb{N}$, and for a nondecreasing sequence of events $\mathbb{P}(A_M) \rightarrow \mathbb{P}(\bigcup_{M \geq 1} A_M) = 1$.

Let f be bounded continuous, hence uniformly continuous on the compact I_ϵ . We cut I_ϵ into disjoint rectangles $R_1, \dots, R_K$ of diameter below the modulus of f at ϵ , so that f varies by at most ϵ on each, and set

$$\hat{f} = \sum_{i \leq K} \mathbf{1}_{R_i} \inf_{R_i} f, \quad \text{so} \quad |\hat{f}(x) - f(x)| \leq \epsilon + \|f\|_\infty \mathbf{1}\{x \notin I_\epsilon\} \text{ for all } x.$$

Proof, the three term bound

Inserting $\hat{f}$ twice and using that it is constant on each R_i , we obtain

$$\begin{aligned} |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| &\leq \mathbb{E}_{X_n}|f - \hat{f}|(X_n) + |\mathbb{E}_{X_n}[\hat{f}(X_n)] - \mathbb{E}_X[\hat{f}(X)]| + \mathbb{E}_X|\hat{f} - f|(X) \\ &\leq 2\epsilon + \|f\|_\infty(\mathbb{P}(X_n \notin I_\epsilon) + \mathbb{P}(X \notin I_\epsilon)) + \sum_{i \leq K} |\Delta_{i,n}| \inf_{R_i} |f| \end{aligned}$$

with $\Delta_{i,n} = \mathbb{P}(X_n \in R_i) - \mathbb{P}(X \in R_i) \rightarrow 0$. Since the last term is a finite sum, it vanishes, while the middle one tends to $2\mathbb{P}(X \notin I_\epsilon) \leq 2\epsilon$. Hence, $\limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \leq 2\epsilon(1 + \|f\|_\infty)$ for every ϵ , which concludes the proof in the continuous case.

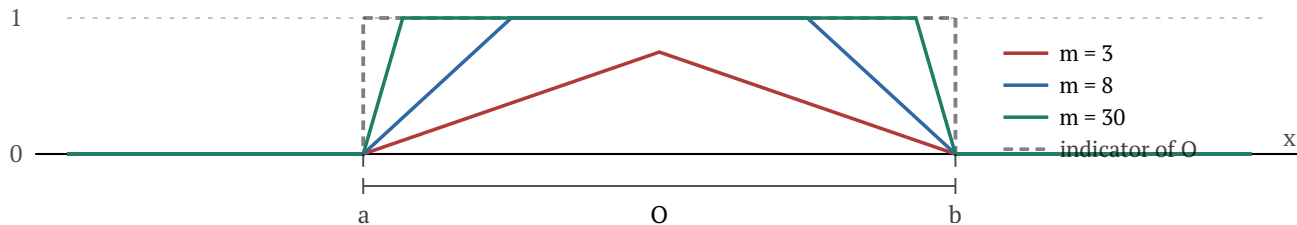
The case of a discontinuous limit. If F_X is not continuous everywhere, the argument is unchanged once the rectangles are well chosen. Indeed, F_X only has countably many discontinuity points, and it suffices to take, in the previous proof, rectangles whose faces avoid these points.

Proof, (ii) to (iii) to (v) to (vi)

(ii) implies (iii). A bounded Lipschitz function is bounded continuous.

(iii) implies (v). Let O be open and $f_m(x) = \min(1, m \text{dist}(x, O^c))$, which is m -Lipschitz, bounded by 1, and satisfies $0 \leq f_m \leq \mathbf{1}_O$ with $f_m \uparrow \mathbf{1}_O$ pointwise, since $\text{dist}(x, O^c) > 0$ exactly on O . Hence, by (iii) and monotone convergence, we obtain

$$\liminf_n \mathbb{P}(X_n \in O) \geq \lim_n \mathbb{E}_{X_n}[f_m(X_n)] = \mathbb{E}_X[f_m(X)] \xrightarrow{m \rightarrow \infty} \mathbb{P}(X \in O)$$



(v) implies (vi). Applying (v) to $O = F^c$ and taking complements, we obtain $\limsup_n \mathbb{P}(X_n \in F) = 1 - \liminf_n \mathbb{P}(X_n \in O) \leq 1 - \mathbb{P}(X \in O) = \mathbb{P}(X \in F)$.

Proof, (v) and (vi) to (vii) and (vii) to (i)

(v) and (vi) imply (vii). Let B be Borel with $\mathbb{P}(X \in \partial B) = 0$. Then

$$\begin{aligned}
 \mathbb{P}(X \in \mathring{B}) &\leq \liminf_n \mathbb{P}(X_n \in \mathring{B}) \leq \liminf_n \mathbb{P}(X_n \in B) && \text{(v) on } \mathring{B}, \text{ then } \mathring{B} \subseteq B \\
 &\leq \limsup_n \mathbb{P}(X_n \in B) \leq \limsup_n \mathbb{P}(X_n \in \bar{B}) && B \subseteq \bar{B} \\
 &\leq \mathbb{P}(X \in \bar{B}) = \mathbb{P}(X \in \mathring{B}) && \text{(vi) on } \bar{B}, \text{ then } \mathbb{P}(X \in \partial B) = 0.
 \end{aligned}$$

Since the chain closes on the value it started from, all its terms are equal.

(vii) implies (i). We take $B = (-\infty, x]$, closed, with $\mathring{B} = (-\infty, x)$ and $\partial B = \{u \leq x : u_j = x_j \text{ for some } j\}$. Since the events $\{X \leq x - 2^{-m} \mathbf{1}\}$ increase to $\{X < x\}$, we have

$$\mathbb{P}(X \in \partial B) = F_X(x) - \mathbb{P}(X < x) = F_X(x) - \lim_{m \rightarrow \infty} F_X(x - 2^{-m} \mathbf{1}) = 0.$$

Hence, by (vii),

$$F_{X_n}(x) = \mathbb{P}(X_n \in B) \rightarrow \mathbb{P}(X \in B) = F_X(x).$$

Proof, equivalence of (ii) and (iv)

(ii) implies (iv). Let $f \geq 0$ be continuous and $M > 0$. Since $f \wedge M$ is bounded continuous and $f \geq f \wedge M$, (ii) and monotone convergence yield

$$\liminf_n \mathbb{E}_{X_n}[f(X_n)] \geq \lim_n \mathbb{E}_{X_n}[(f \wedge M)(X_n)] = \mathbb{E}_X[(f \wedge M)(X)] \xrightarrow{M \rightarrow \infty} \mathbb{E}_X[f(X)]$$

(iv) implies (ii). Let f be bounded continuous. Since both $\|f\|_\infty + f$ and $\|f\|_\infty - f$ are continuous and nonnegative, applying (iv) to each of them yields

$$\liminf_n \mathbb{E}_{X_n}[f(X_n)] \geq \mathbb{E}_X[f(X)] \quad \text{and} \quad \limsup_n \mathbb{E}_{X_n}[f(X_n)] \leq \mathbb{E}_X[f(X)].$$

Exercise, non-convergence of differences

Claim. $X_n \xrightarrow{d} X$ does not imply $X_n - X \xrightarrow{d} 0$.

Solution. Let $U \sim \text{Unif}([0, 1])$, $X_n = U$ for every n , and $X = 1 - U$. Then $X \sim \text{Unif}([0, 1])$, so that $X_n \stackrel{d}{=} X$ and, in particular, $X_n \xrightarrow{d} X$. However,

$$X_n - X = 2U - 1 \sim \text{Unif}([-1, 1]),$$

which does not converge in distribution to 0.

Origin of the failure. Convergence in distribution controls the law of X_n and the law of X separately, but never the joint law of (X_n, X) . Similarly, $X \stackrel{d}{=} Y$ yields no information at all about (X, Z) and (Y, Z) , which is the reason why Slutsky requires one of the two sequences to have a constant limit.

Exercise, Dirac masses

Claim. $\delta_{x_n} \xrightarrow{d} \delta_x$ if and only if $x_n \rightarrow x$, for $x_n, x \in \mathbb{R}^d$.

The easy direction. If $x_n \rightarrow x$, then for every bounded continuous f , we have

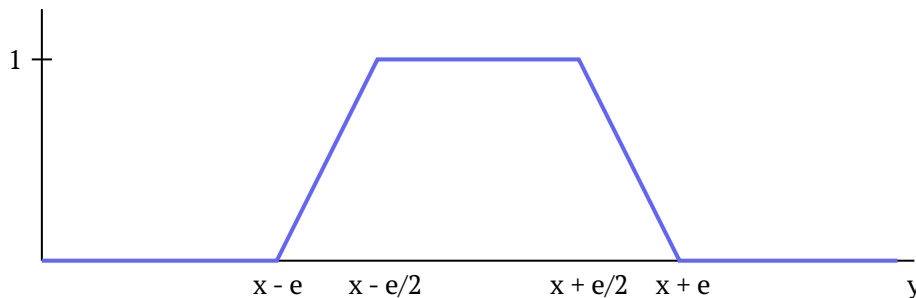
$$\mathbb{E}_{X \sim \delta_{x_n}}[f(X)] = f(x_n) \rightarrow f(x) = \mathbb{E}_{X \sim \delta_x}[f(X)]$$

by continuity of f , so that portmanteau (ii) applies.

The other direction requires a test function that isolates x . Since it must be bounded and continuous to be admissible, an indicator does not work.

Exercise, the isolating test function

For $\epsilon > 0$, we consider $f_\epsilon(y) = \min(1, (2 - \frac{2}{\epsilon}\|y - x\|)_+)$, which is continuous, bounded by 1, equal to 1 on $B(x, \epsilon/2)$ and to 0 outside $B(x, \epsilon)$.



If $\delta_{x_n} \xrightarrow{d} \delta_x$, then portmanteau (ii) yields $f_\epsilon(x_n) \rightarrow f_\epsilon(x) = 1$, so that $f_\epsilon(x_n) > 0$ for n large, and hence $\|x_n - x\| < \epsilon$ for n large. Since this holds for every $\epsilon > 0$, we obtain $x_n \rightarrow x$.

Exercise, strictness of the inequalities

Claim. A single sequence shows that the inequalities in (v) and (vi) can be strict, and that the boundary hypothesis in (vii) cannot be dropped.

The sequence. Let $X_n \sim \mathcal{N}(0, 1/n)$ and $X \equiv 0$. For $x < 0$, $\mathbb{P}(X_n \leq x) = \mathbb{P}(\mathcal{N}(0, 1) \leq x\sqrt{n}) \rightarrow 0$, and for $x > 0$ it tends to 1. Since every $x \neq 0$ is a continuity point of F_X , while 0 is not, we obtain $X_n \xrightarrow{d} X$.

(v) is strict on $O = (0, \infty)$. By symmetry, $\mathbb{P}(X_n \in O) = \frac{1}{2}$ for every n , while $\mathbb{P}(X \in O) = 0$. Hence, the inequality $\liminf_n \frac{1}{2} \geq 0$ holds with a gap of $\frac{1}{2}$.

(vi) is strict on $F = \{0\}$. Since each X_n has a density, $\mathbb{P}(X_n \in F) = 0$, while $\mathbb{P}(X \in F) = 1$. Again, the inequality holds and is strict, now in the other direction.

(vii) fails on $B = (0, \infty)$. Its boundary is $\{0\}$, which carries all of the mass of X , so that the hypothesis of (vii) is violated. The conclusion fails as well, since $\mathbb{P}(X_n \in B) = \frac{1}{2}$ does not tend to $\mathbb{P}(X \in B) = 0$.

Exercise, the maximum of uniforms

Claim. Let $U_1, U_2, \dots$ be independent and uniform on $[0, 1]$ and $M_n = \max_{i \leq n} U_i$. Then $n(1 - M_n) \xrightarrow{d} \mathcal{E}(1)$.

Solution. The maximum has an explicit distribution function. Indeed, since the U_i are independent, we have for $0 \leq t \leq n$

$$\mathbb{P}(n(1 - M_n) > t) = \mathbb{P}\left(M_n < 1 - \frac{t}{n}\right) = \prod_{i \leq n} \mathbb{P}\left(U_i < 1 - \frac{t}{n}\right) = \left(1 - \frac{t}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-t},$$

so that $\mathbb{P}(n(1 - M_n) \leq t) \rightarrow 1 - e^{-t}$ for every $t > 0$. Since the limit is the exponential distribution function, which is continuous everywhere, no point is excluded and the convergence in law holds.

Two remarks. The scaling is n and not $\sqrt{n}$, since a maximum is not an average and the central limit theorem does not apply to it. Furthermore, since $\mathbb{P}(M_n \leq 1 - \epsilon) = (1 - \epsilon)^n \rightarrow 0$, we have $M_n \rightarrow 1$ in probability, and the claim yields the rate of this convergence.

Exercise, convergence on a dense set

Claim. Let F be a distribution function and $D \subseteq \mathbb{R}$ dense. If $F_{X_n}(x) \rightarrow F(x)$ for every $x \in D$, then $X_n \xrightarrow{d} X$ with $F_X = F$.

Solution. Let x be a continuity point of F and let us take $u < x < v$ with $u, v \in D$, which is possible by density. By monotonicity of each F_{X_n} , we have $F_{X_n}(u) \leq F_{X_n}(x) \leq F_{X_n}(v)$, and letting $n \rightarrow \infty$, we obtain

$$F(u) \leq \liminf_n F_{X_n}(x) \leq \limsup_n F_{X_n}(x) \leq F(v).$$

Eventually, letting $u \uparrow x$ and $v \downarrow x$ inside D , both ends tend to $F(x)$ since F is continuous at x , which concludes the proof.

Use in practice. This is how we check convergence in law in practice. Indeed, it suffices to verify the limit on the rationals, or on any convenient countable set, and the continuity points come for free. The argument is again the squeeze of the portmanteau proof, with a dense set in place of a sequence.

Part III. Characteristic functions

The Fourier transform of a law

Definition. Let X be a random vector of $\mathbb{R}^d$. Its characteristic function, the Fourier transform of its law, is

$$\phi_X : \mathbb{R}^d \rightarrow \mathbb{C}, \quad \phi_X(t) = \mathbb{E}_X[e^{i\langle t, X \rangle}] = \int_{\mathbb{R}^d} e^{i\langle t, x \rangle} d\mathbb{P}_X(x).$$

Note that it is defined for every t and every X , with no integrability assumption, since $|e^{i\langle t, x \rangle}| = 1$.

Motivation. Convergence in distribution is a statement about a family of test functions indexed by all of $C_b(\mathbb{R}^d)$, and we show in this part and the next that the family $x \mapsto e^{i\langle t, x \rangle}$ already suffices. Furthermore, the Fourier transform enjoys many convenient properties.

Elementary properties

Proposition. Let X, Y be random vectors of $\mathbb{R}^d$.

(i) $\phi_X = \phi_Y \iff \mathbb{P}_X = \mathbb{P}_Y \iff X \stackrel{d}{=} Y$.

(ii) $|\phi_X(t)| \leq 1$ for every t , and $\phi_X(0) = 1$.

(iii) ϕ_X is uniformly continuous on $\mathbb{R}^d$.

(iv) For $A \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^d$, $\phi_{AX+b}(t) = \phi_X(A^\top t) e^{i\langle b, t \rangle}$.

(v) $\phi_{-X}(t) = \phi_X(-t) = \overline{\phi_X(t)}$, so X is symmetric if and only if ϕ_X is real valued.

Proof of (i), the smoothing identity

(i) The implication $\Leftarrow$ is immediate. For the converse, we rely on the following lemma.

Lemma. Let $Z \sim \mathcal{N}(0, I_d)$ be independent of X , let $\sigma > 0$, and let q_σ be the density of $\mathcal{N}(0, \sigma^2 I_d)$. Then $X + \sigma Z$ has the density

$$p_\sigma^X(x) = \mathbb{E}_X[q_\sigma(x - X)] = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) e^{-\sigma^2 \|t\|^2/2} dt.$$

$$\begin{aligned} \mathbb{E}_X[q_\sigma(x - X)] &= \int_{\mathbb{R}^d} q_\sigma(x - y) d\mathbb{P}_X(y) && \text{definition of a convolution} \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (2\pi)^{-d} e^{i\langle t, y-x \rangle} e^{-\sigma^2 \|t\|^2/2} dt d\mathbb{P}_X(y) && \text{gaussian transform, next part} \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} (2\pi)^{-d} e^{-i\langle t, x \rangle} e^{i\langle t, y \rangle} e^{-\sigma^2 \|t\|^2/2} d\mathbb{P}_X(y) dt && \text{Fubini, modulus } e^{-\sigma^2 \|t\|^2/2} \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) e^{-\sigma^2 \|t\|^2/2} dt && \text{definition of } \phi_X. \quad \blacksquare \end{aligned}$$

Proof of (i), removing the smoothing

Let us suppose that $\phi_X = \phi_Y$. By the lemma, $X + \sigma Z$ and $Y + \sigma Z$ have the same density for every $\sigma > 0$, hence the same law.

Let f be bounded continuous, and let us take $\sigma_n = 1/n$, since dominated convergence is a statement about sequences.

For almost every ω , the vector $Z(\omega)$ is finite, so that $X(\omega) + \sigma_n Z(\omega) \rightarrow X(\omega)$, and since f is continuous, $f(X + \sigma_n Z) \rightarrow f(X)$ almost surely.

Furthermore, $|f(X + \sigma_n Z)| \leq \|f\|_\infty$ for every n , and the constant $\|f\|_\infty$ is integrable since $\mathbb{P}$ is a probability measure. Since both facts hold with Y in place of X , dominated convergence yields

$$\mathbb{E}_X[f(X)] = \lim_{n \rightarrow \infty} \mathbb{E}_{X,Z}[f(X + \sigma_n Z)] = \lim_{n \rightarrow \infty} \mathbb{E}_{Y,Z}[f(Y + \sigma_n Z)] = \mathbb{E}_Y[f(Y)],$$

where the middle equality holds term by term by the first step.

Proof of (i), separating the laws

We now show that bounded continuous functions determine the law.

For F closed and nonempty, let $f_m(x) = (1 - m \operatorname{dist}(x, F))_+$ for $m \in \mathbb{N}$. Each f_m is bounded by 1, is m -Lipschitz, equals 1 on F , and decreases pointwise to $\mathbf{1}_F$. Indeed, if $x \notin F$, then $\operatorname{dist}(x, F) > 0$ since F is closed, so $f_m(x) = 0$ as soon as $m > 1/\operatorname{dist}(x, F)$.

Dominated convergence once more, with the constant 1 as dominating function, yields

$$\mathbb{P}(X \in F) = \lim_{m \rightarrow \infty} \mathbb{E}_X[f_m(X)] = \lim_{m \rightarrow \infty} \mathbb{E}_Y[f_m(Y)] = \mathbb{P}(Y \in F).$$

Since closed sets form a π -system generating the Borel σ -algebra, two probability measures agreeing on all of them are equal, hence $\mathbb{P}_X = \mathbb{P}_Y$, which concludes the proof. ■

Proof of the elementary properties

(ii) $\phi_X(0) = \mathbb{E}_X[1] = 1$ and $|\phi_X(t)| = |\mathbb{E}_X[e^{i\langle t, X \rangle}]| \leq \mathbb{E}_X[|e^{i\langle t, X \rangle}|] = 1$.

(iii) Given $\epsilon > 0$, we pick M with $\mathbb{P}(X \notin B(0, M)) \leq \epsilon$ and split the expectation at $B(0, M)$. Outside, $|e^{ia} - e^{ib}| \leq 2$ and the mass is at most ϵ , while inside, $|e^{ia} - e^{ib}| \leq |a - b|$ (mean value inequality) and $|\langle t_1 - t_2, x \rangle| \leq M\|t_1 - t_2\|$ by Cauchy and Schwarz. Hence

$$|\phi_X(t_1) - \phi_X(t_2)| \leq \int_{B(0, M)} |e^{i\langle t_1, x \rangle} - e^{i\langle t_2, x \rangle}| d\mathbb{P}_X(x) + 2\epsilon \leq M\|t_1 - t_2\| + 2\epsilon.$$

So, for every pair with $\|t_1 - t_2\| \leq \epsilon/M$, we obtain $|\phi_X(t_1) - \phi_X(t_2)| \leq 3\epsilon$.

(iv) and (v) are direct consequences of $\langle t, Ax + b \rangle = \langle A^\top t, x \rangle + \langle b, t \rangle$ and $\overline{e^{iu}} = e^{-iu}$.

Further properties of characteristic functions

Proposition, continued. With $X^\alpha = \prod_{j \leq d} X_j^{\alpha_j}$ for a multi-index α and $|\alpha| = \sum_j \alpha_j$,

(vi) if $\mathbb{E}_X[\|X\|^r] < \infty$ then $\phi_X \in C^r$, and $\partial^\alpha \phi_X(t) = i^{|\alpha|} \mathbb{E}_X[X^\alpha e^{i\langle t, X \rangle}]$ for $|\alpha| \leq r$.

(vii) the converse is false, but if ϕ_X has a derivative of even order $2r$ at 0 then $\mathbb{E}_X[\|X\|^{2r}] < \infty$.

(viii) if X has a density with respect to Lebesgue measure then $\phi_X(t) \rightarrow 0$ as $\|t\| \rightarrow \infty$.

(ix) if $\phi_X \in L^1$ then X has a bounded continuous density, given by the inversion formula $p(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) dt$.

(x) X and Y are independent if and only if $\phi_{(X,Y)}(s, t) = \phi_X(s)\phi_Y(t)$ for all (s, t) .

(xi) if X and Y are independent then $\mathbb{P}_{X+Y} = \mathbb{P}_X * \mathbb{P}_Y$ and $\phi_{X+Y} = \phi_X \phi_Y$.

Proof, differentiability and moments

(vi) We differentiate under the integral sign. The candidate derivative $x \mapsto ix_j e^{i\langle t, x \rangle}$ is dominated, uniformly in t , by $|x_j| \leq \|x\|$, which is integrable when $r \geq 1$. We then iterate r times, the α -th derivative being dominated by $\|x\|^{|\alpha|}$.

(vii) Let us take $d = 1$ and $r = 1$. If $\phi_X''(0)$ exists, then it equals the limit of the symmetric second difference, and since $\phi_X(t) + \phi_X(-t) = 2\mathbb{E}_X[\cos(tX)]$, we obtain

$$\phi_X''(0) = \lim_{t \rightarrow 0} \frac{\phi_X(t) + \phi_X(-t) - 2\phi_X(0)}{t^2} = -2 \lim_{t \rightarrow 0} \mathbb{E}_X \left[\frac{1 - \cos(tX)}{t^2} \right].$$

Since $(1 - \cos(tx))/t^2 \rightarrow x^2/2$ as $t \rightarrow 0$ and the integrand is nonnegative, Fatou's lemma along any sequence $t_n \rightarrow 0$ yields

$$\mathbb{E}_X[X^2] = \mathbb{E}_X \left[\liminf_n 2 \frac{1 - \cos(t_n X)}{t_n^2} \right] \leq 2 \liminf_n \mathbb{E}_X \left[\frac{1 - \cos(t_n X)}{t_n^2} \right] = -\phi_X''(0) < \infty.$$

The same reasoning applies to higher orders.

(viii), The Riemann and Lebesgue lemma

(viii) Item (viii) is a statement about integrable functions rather than about probability, and a density is such a function.

Lemma (Riemann and Lebesgue). For $f \in L^1(\mathbb{R}^d)$ set $\mathcal{F}f(t) = \int_{\mathbb{R}^d} f(x) e^{i\langle t, x \rangle} dx$. Then $\mathcal{F}f(t) \rightarrow 0$ as $\|t\| \rightarrow \infty$.

Proof. Let $t \neq 0$ and $h = \pi t / \|t\|^2$, so that $\langle t, h \rangle = \pi$ and $\|h\| = \pi / \|t\|$. Substituting $x = y + h$ and using $e^{i\pi} = -1$, we obtain

$$\mathcal{F}f(t) = \int f(x) e^{i\langle t, x \rangle} dx = e^{i\langle t, h \rangle} \int f(y + h) e^{i\langle t, y \rangle} dy = - \int f(y + h) e^{i\langle t, y \rangle} dy.$$

Averaging this with the original expression cancels the oscillating factor, and we obtain

$$2\mathcal{F}f(t) = \int (f(y) - f(y + h)) e^{i\langle t, y \rangle} dy, \quad |\mathcal{F}f(t)| \leq \frac{1}{2} \|f - f(\cdot + h)\|_{L^1}.$$

Proof of the lemma, translation in L^1

Since $\|h\| = \pi/\|t\|$, it remains to prove that $\|f - f(\cdot + h)\|_{L^1} \rightarrow 0$ as $h \rightarrow 0$.

For g continuous with compact support, all the functions $g(\cdot + h)$ with $\|h\| \leq 1$ vanish off one fixed compact set K , and since g is uniformly continuous, we have $\|g - g(\cdot + h)\|_{L^1} \leq \lambda(K) \sup_{x \in \mathbb{R}^d} |g(x) - g(x + h)| \rightarrow 0$, where λ denotes the Lebesgue measure.

For a general f , we use that such g are dense in $L^1(\mathbb{R}^d)$ (see Rudin, *Real and Complex Analysis*, Theorem 3.14). Given $\epsilon > 0$, we pick g with $\|f - g\|_{L^1} < \epsilon/3$, and since translation preserves the L^1 norm, we obtain

$$\|f - f(\cdot + h)\|_{L^1} \leq 2\|f - g\|_{L^1} + \|g - g(\cdot + h)\|_{L^1} < \epsilon$$

for $\|h\|$ small enough. ■

Proof of (viii). If X has a density p with respect to Lebesgue measure, then $p \in L^1(\mathbb{R}^d)$ and $\phi_X = \mathcal{F}p$, so that $\phi_X(t) \rightarrow 0$ as $\|t\| \rightarrow \infty$. ■

(ix), The Fourier inversion theorem

(ix) Theorem (Fourier inversion). Let X be a random vector of $\mathbb{R}^d$ with $\phi_X \in L^1(\mathbb{R}^d)$. Then $\mathbb{P}_X$ has a bounded continuous density with respect to Lebesgue measure, namely

$$p(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi_X(t) dt.$$

We write $\phi = \phi_X$. Since Fubini does not apply directly to the inversion integral, we first smooth it by setting, for $\sigma > 0$,

$$p_\sigma(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi(t) e^{-\sigma^2 \|t\|^2/2} dt = \mathbb{E}_X[q_\sigma(x - X)],$$

where the second equality is the lemma proved for (i), with $Z \sim \mathcal{N}(0, I_d)$ independent of X and q_σ the density of $\mathcal{N}(0, \sigma^2 I_d)$. In particular, p_σ is the density of $X + \sigma Z$.

Proof of the theorem, the transform side

We first show that $p_\sigma \rightarrow p$ uniformly on $\mathbb{R}^d$.

$$\begin{aligned} |p_\sigma(x) - p(x)| &= (2\pi)^{-d} \left| \int_{\mathbb{R}^d} e^{-i\langle t, x \rangle} \phi(t) \left(e^{-\sigma^2 \|t\|^2/2} - 1 \right) dt \right| && \text{definitions of } p_\sigma \text{ and } p \\ &\leq (2\pi)^{-d} \int_{\mathbb{R}^d} |\phi(t)| \left(1 - e^{-\sigma^2 \|t\|^2/2} \right) dt && \text{triangle inequality} \\ &\xrightarrow[\sigma \rightarrow 0]{} 0 && \text{dominated by } |\phi| \in L^1 \end{aligned}$$

Since the last bound does not depend on x , the convergence is uniform. Furthermore, the same domination yields $\|p\|_\infty \leq (2\pi)^{-d} \|\phi\|_{L^1}$ and, by dominated convergence in x , the continuity of p .

Proof of the theorem, the law side

Given a bounded rectangle $R = (a, b]$ with $\mathbb{P}(X \in \partial R) = 0$, we have

$$\begin{aligned}\mathbb{P}(X \in R) &= \lim_{\sigma \rightarrow 0} \mathbb{P}(X + \sigma Z \in R) && \sigma Z \rightarrow 0 \text{ a.s. and } \mathbb{P}(X \in \partial R) = 0 \\ &= \lim_{\sigma \rightarrow 0} \int_R p_\sigma(x) dx && p_\sigma \text{ is the density of } X + \sigma Z \\ &= \int_R p(x) dx && p_\sigma \rightarrow p \text{ uniformly, } R \text{ bounded}\end{aligned}$$

Such rectangles form a π -system generating the Borel σ -algebra, since for each coordinate all but countably many levels carry no mass. As two probability measures agreeing on it are equal, p is a density of $\mathbb{P}_X$, which concludes the proof. ■

Corollary, inversion in L^1 . If $f \in L^1(\mathbb{R}^d)$ is a probability density with $\mathcal{F}f \in L^1(\mathbb{R}^d)$, then $f = p$ almost everywhere. In other words, a density whose transform is integrable has a bounded continuous version.

Proof, independence and sums

(x) If X and Y are independent, then by Fubini we have $\mathbb{E}_{X,Y}[e^{i\langle s,X\rangle+i\langle t,Y\rangle}] = \phi_X(s)\phi_Y(t)$. Conversely, since the right side is the characteristic function of $\mathbb{P}_X \otimes \mathbb{P}_Y$, (i) shows that the joint law equals the product law, which is independence.

(xi) We take $s = t$ in (x), or write directly $\mathbb{E}_{X,Y}[e^{i\langle t,X+Y\rangle}] = \mathbb{E}_X[e^{i\langle t,X\rangle}]\mathbb{E}_Y[e^{i\langle t,Y\rangle}]$ by Fubini.

The standard gaussian, by a differential equation

Proposition. Let $X \sim \mathcal{N}(0, 1)$ on $\mathbb{R}$. Then $\phi_X(t) = e^{-t^2/2}$ for every $t \in \mathbb{R}$, so the standard gaussian is its own Fourier transform up to the normalising constant.

We give three proofs: a differential equation here, analytic continuation and a contour integral next. In each of them, we compute $\phi_X(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{itx} e^{-x^2/2} dx$, and we show here that it solves $\phi'_X = -t\phi_X$, whose solutions are the multiples of $e^{-t^2/2}$. By (vi) with $r = 1$, we have

$$\begin{aligned}\phi'_X(t) &= \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} \underbrace{e^{itx}}_{\text{easy to differentiate}} \underbrace{xe^{-x^2/2}}_{\text{easy to integrate}} dx && \text{differentiation under the integral} \\ &= \frac{-i}{\sqrt{2\pi}} \int_{\mathbb{R}} ite^{itx} e^{-x^2/2} dx = -t\phi_X(t) && \text{integration by parts, boundary terms vanish}\end{aligned}$$

Thus $\phi'_X + t\phi_X = 0$, hence $\phi_X(t) = \phi_X(0)e^{-t^2/2}$, and since $\phi_X(0) = 1$ by (ii), we obtain

$$\phi_X(t) = e^{-t^2/2}.$$

The standard gaussian, by analytic continuation

We set $F(z) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{zx} e^{-x^2/2} dx$ for $z \in \mathbb{C}$, so that $\phi_X(t) = F(it)$. Since the integrand is holomorphic in z and locally dominated by $e^{R|x|} e^{-x^2/2}$ on $\{|z| \leq R\}$, the function F is entire.

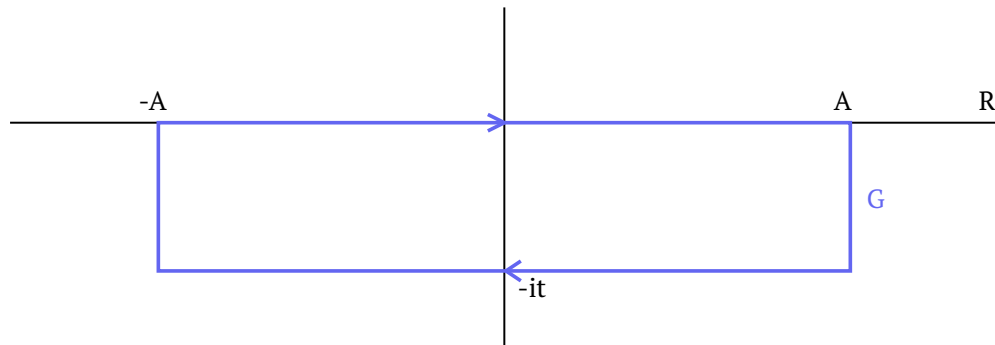
For $z = s$ real, completing the square, $sx - \frac{x^2}{2} = -\frac{(x-s)^2}{2} + \frac{s^2}{2}$, and using the gaussian integral, we obtain

$$F(s) = \frac{e^{s^2/2}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-s)^2/2} dx = e^{s^2/2}.$$

The entire functions F and $z \mapsto e^{z^2/2}$ agree on $\mathbb{R}$, which has an accumulation point, so they agree on $\mathbb{C}$ by the identity theorem. Hence, $\phi_X(t) = F(it) = e^{-t^2/2}$.

The standard gaussian, by a contour integral

We complete the square inside the exponential, $itx - \frac{x^2}{2} = -\frac{(x-it)^2}{2} - \frac{t^2}{2}$, so that it suffices to prove $\int_{\mathbb{R}} e^{-(x-it)^2/2} dx = \sqrt{2\pi}$. Let us take $t > 0$, the case $t < 0$ following from property (v), and apply Cauchy's theorem to the entire function $g(u) = e^{-u^2/2}$ on the rectangle Γ_A with vertices $\pm A$ and $\pm A - it$.



The two vertical sides have length $|t|$ and integrand bounded by $e^{-(A^2-t^2)/2}$, so they vanish as $A \rightarrow \infty$. The top side tends to $\int_{\mathbb{R}} e^{-x^2/2} dx = \sqrt{2\pi}$ and the bottom to $\int_{\mathbb{R}} e^{-(x-it)^2/2} dx$, both by dominated convergence. Since the total is 0, the two limits are equal.

The gaussian in $\mathbb{R}^d$

Isotropic case. If $X \sim \mathcal{N}(0, I_d)$, the coordinates are independent standard gaussians, so that (x) and the scalar computation yield

$$\phi_X(t) = \prod_{j \leq d} \phi_{X_j}(t_j) = \prod_{j \leq d} e^{-t_j^2/2} = e^{-\|t\|^2/2}.$$

General case. Given $\Sigma \succeq 0$, we diagonalise $\Sigma = P^\top D P$ with P orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_d)$, and set $\Sigma^{1/2} = P^\top D^{1/2} P$. If $X' \sim \mathcal{N}(0, I_d)$, then $X = \Sigma^{1/2} X' + \mu \sim \mathcal{N}(\mu, \Sigma)$, and (iv) yields

$$\phi_X(t) = e^{i\langle \mu, t \rangle} e^{-\|\Sigma^{1/2} t\|^2/2} = \exp(i\langle \mu, t \rangle - \tfrac{1}{2} t^\top \Sigma t).$$

Theorem. Let $\mu \in \mathbb{R}^d$ and let Σ be symmetric positive semidefinite, possibly singular. Then $X \sim \mathcal{N}(\mu, \Sigma)$ if and only if

$$\phi_X(t) = \exp \left(i\langle \mu, t \rangle - \tfrac{1}{2} t^\top \Sigma t \right), \quad t \in \mathbb{R}^d.$$

Part IV. Tightness, Prohorov, Levy

Uniform tightness

Definition. A family $(X_a)_{a \in A}$ of random vectors of $\mathbb{R}^d$ is uniformly tight if

$$\forall \epsilon > 0, \exists M > 0 \text{ such that } \sup_{a \in A} \mathbb{P}(\|X_a\| \geq M) \leq \epsilon,$$

equivalently

$$\sup_{a \in A} \mathbb{P}(\|X_a\| \geq M) \xrightarrow{M \rightarrow \infty} 0.$$

A single random vector is tight, since $\{\|X\| \geq M\} \downarrow \emptyset$ as $M \uparrow \infty$, so that $\mathbb{P}(\|X\| \geq M) \rightarrow 0$ by monotone convergence. Similarly, a finite family is tight, by taking the largest of the finitely many M .

Reading. Tightness means that no mass escapes to infinity. It is a property of the laws $\mathbb{P}_{X_a}$ rather than of the random vectors, and it can be seen as the analogue of boundedness for a sequence of real numbers.

Prohorov's theorem

Theorem. Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random vectors of $\mathbb{R}^d$.

1. If $X_n \xrightarrow{d} X$ for some random vector X , then $(X_n)_n$ is uniformly tight.
2. If $(X_n)_n$ is uniformly tight then there exist a random vector X and a subsequence $(X_{\varphi(n)})$ with $X_{\varphi(n)} \xrightarrow{d} X$.

The analogy. In $\mathbb{R}$, a convergent sequence is bounded, and from a bounded sequence we can extract a convergent subsequence. Similarly, a sequence convergent in distribution is tight, and from a tight sequence we can extract a subsequence convergent in distribution. This allows us to use compactness arguments.

Proof of Prohorov, first item

Let $\epsilon > 0$. Since X is tight, there is M with $\mathbb{P}(\|X\| \geq M) < \epsilon$, and since the set $\{x : \|x\| \geq M\}$ is closed, portmanteau (vi) yields

$$\limsup_n \mathbb{P}(\|X_n\| \geq M) \leq \mathbb{P}(\|X\| \geq M) < \epsilon.$$

Hence, there is N with $\mathbb{P}(\|X_n\| \geq M) < 2\epsilon$ for every $n \geq N$. We handle the finitely many indices $n < N$ by enlarging M , since each of them is tight on its own. Thus $\sup_n \mathbb{P}(\|X_n\| \geq M') < 2\epsilon$ for some $M' \geq M$, and ϵ was arbitrary.

Remark. The two steps mirror the proof that a convergent sequence is bounded, in which the tail is controlled by the limit and the head is finite.

Helly's selection lemma

Lemma (Helly). From any sequence (F_n) of cumulative distribution functions on $\mathbb{R}^d$ one can extract a subsequence converging, at every continuity point, to a nondecreasing right continuous $F : \mathbb{R}^d \rightarrow [0, 1]$.

Proof. Diagonal extraction. Since $\mathbb{Q}^d$ is countable, we enumerate it as $q^{(1)}, q^{(2)}, \dots$. The sequence $(F_n(q^{(1)}))_n$ lies in $[0, 1]$, hence has a convergent subsequence, along which we extract again for $q^{(2)}$, and so on. The diagonal subsequence φ , whose n -th term is the n -th term of the n -th extraction, satisfies $F_{\varphi(n)}(q) \rightarrow G(q)$ for every $q \in \mathbb{Q}^d$, and G is nondecreasing as a pointwise limit of nondecreasing functions.

The candidate limit. We write $q > x$ when $q_j > x_j$ for every coordinate j , and we set

$$F(x) = \inf \{ G(q) : q \in \mathbb{Q}^d, q > x \} \in [0, 1].$$

If $x \leq y$, then $\{q > y\} \subset \{q > x\}$, and since an infimum over a smaller set is larger, $F(x) \leq F(y)$.

Proof of Helly, right continuity and convergence

Right continuity. Let $x^{(m)} \downarrow x$ coordinatewise. By monotonicity, $F(x^{(m)}) \downarrow L \geq F(x)$. Given $\epsilon > 0$, by definition of the infimum, we pick $q \in \mathbb{Q}^d$ with $q > x$ and $G(q) \leq F(x) + \epsilon$. For m large, $x^{(m)} < q$, so that q competes in the infimum defining $F(x^{(m)})$ and $F(x^{(m)}) \leq G(q) \leq F(x) + \epsilon$. Hence, $L = F(x)$.

Convergence at a continuity point x . Let $\epsilon > 0$. By continuity of F at x , there is $y < x$ with $F(y) \geq F(x) - \epsilon$. We pick $r \in \mathbb{Q}^d$ with $y < r < x$ and $q \in \mathbb{Q}^d$ with $q > x$ and $G(q) \leq F(x) + \epsilon$. Then

$$\begin{aligned} \liminf_n F_{\varphi(n)}(x) &\geq \lim_n F_{\varphi(n)}(r) = G(r) \geq F(y) \geq F(x) - \epsilon & r < x, r > y \\ \limsup_n F_{\varphi(n)}(x) &\leq \lim_n F_{\varphi(n)}(q) = G(q) \leq F(x) + \epsilon & x < q \end{aligned}$$

so that $F_{\varphi(n)}(x) \rightarrow F(x)$. ■

Proof of Prohorov, second item

We apply Helly to $F_n = F_{X_n}$ and denote by F the limit along φ .

The limit may be defective, in the sense that F need not tend to 1 as all coordinates tend to $+\infty$ (i.e. mass can escape). Tightness forbids this. Indeed, given $\epsilon > 0$, we pick M with $\sup_n \mathbb{P}(\|X_n\| \geq M) \leq \epsilon$. Then, along the extracted subsequence and at continuity points,

$$x_j > M \ \forall j \implies F(x) = \lim_n F_n(x) \geq 1 - \epsilon, \quad x_j < -M \text{ for some } j \implies F(x) \leq \epsilon.$$

Hence, F tends to 1 and to 0 at the two ends and is a cumulative distribution function, namely that of some random vector X with $X_{\varphi(n)} \xrightarrow{d} X$.

Levy's continuity theorem

Theorem. Let (X_n) and X be random vectors of $\mathbb{R}^d$.

1. If $X_n \xrightarrow{d} X$ then $\phi_{X_n}(t) \rightarrow \phi_X(t)$ for every $t \in \mathbb{R}^d$.
2. If there is $\phi : \mathbb{R}^d \rightarrow \mathbb{C}$, continuous at 0, with $\phi_{X_n}(t) \rightarrow \phi(t)$ for every t , then ϕ is the characteristic function of some random vector X and $X_n \xrightarrow{d} X$.

Continuity at zero. In item 2, the limit ϕ is only assumed to be continuous at the single point 0, rather than to be a characteristic function. Without that assumption, the conclusion is false. Indeed, for $X_n \sim \mathcal{N}(0, n)$, $\phi_{X_n}(t) = e^{-nt^2/2} \rightarrow \mathbf{1}\{t = 0\}$ pointwise, a limit discontinuous at 0, and X_n does not converge in distribution.

Proof of Levy, first item

The functions $x \mapsto \cos\langle t, x \rangle$ and $x \mapsto \sin\langle t, x \rangle$ are bounded and continuous, so that portmanteau (ii) applies to each of them and yields

$$\phi_{X_n}(t) = \mathbb{E}_{X_n}[\cos\langle t, X_n \rangle] + i\mathbb{E}_{X_n}[\sin\langle t, X_n \rangle] \rightarrow \mathbb{E}_X[\cos\langle t, X \rangle] + i\mathbb{E}_X[\sin\langle t, X \rangle] = \phi_X(t).$$

Proof of Levy, second item, at most one limit point

Let us suppose that (X_n) is uniformly tight and that two subsequences converge in distribution, $X_{\varphi_1(n)} \xrightarrow{d} X$ and $X_{\varphi_2(n)} \xrightarrow{d} X'$. By item 1,

$$\phi_X(t) = \lim_n \phi_{X_{\varphi_1(n)}}(t) = \phi(t) = \lim_n \phi_{X_{\varphi_2(n)}}(t) = \phi_{X'}(t) \quad \text{for every } t,$$

so that $X \stackrel{d}{=} X'$ by injectivity (property (i)).

A tight sequence with a single limit point converges. Indeed, if $X_n \not\xrightarrow{d} X$, there are $\epsilon > 0$, a bounded continuous f and a subsequence along which $|\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \geq \epsilon$. Since that subsequence is tight, Prohorov yields a further subsequence converging in distribution, necessarily to X by the paragraph above, which contradicts the separation.

Hence, it only remains to show that (X_n) is uniformly tight in order to prove item 2.

Proof of Levy, tightness from continuity at zero

Key inequality. For a real random variable Z and $\delta > 0$, we have by Fubini and $\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - e^{itz}) dt = 2 - \frac{2 \sin(\delta z)}{\delta z}$,

$$\frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \phi_Z(t)) dt = 2 \mathbb{E}_Z \left[1 - \frac{\sin(\delta Z)}{\delta Z} \right] \geq \mathbb{P}(|Z| \geq 2/\delta),$$

since $1 - \frac{\sin u}{u} \geq 0$ for all u , and $1 - \frac{\sin u}{u} \geq \frac{1}{2}$ as soon as $|u| \geq 2$.

Let us apply this to $Z = \langle e_j, X_n \rangle$, whose characteristic function is $s \mapsto \phi_{X_n}(se_j)$. Since the integrand is bounded by 2, dominated convergence yields

$$\limsup_n \mathbb{P}(|X_{n,j}| \geq 2/\delta) \leq \frac{1}{\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \phi(se_j)) ds,$$

and the right-hand side tends to 0 as $\delta \rightarrow 0$ since ϕ is continuous at 0 with $\phi(0) = \lim_n \phi_{X_n}(0) = 1$.

Hence, each coordinate is tight, so that (X_n) is tight and the previous slide concludes.

The Cramer and Wold device

Corollary. $X_n \xrightarrow{d} X$ in $\mathbb{R}^d$ if and only if $\langle t, X_n \rangle \xrightarrow{d} \langle t, X \rangle$ in $\mathbb{R}$ for every $t \in \mathbb{R}^d$.

Proof. For $s \in \mathbb{R}$, we have $\phi_{\langle t, X_n \rangle}(s) = \mathbb{E}_{X_n}[e^{is\langle t, X_n \rangle}] = \phi_{X_n}(st)$. If all one-dimensional projections converge in distribution, then taking $s = 1$ yields $\phi_{X_n}(t) \rightarrow \phi_X(t)$ for every t , and since ϕ_X is continuous, Levy item 2 applies. The other implication follows from Levy item 1 read at st .

Use of the device. It reduces every multivariate limit theorem to its scalar version, and we prove the multivariate central limit theorem this way in a few slides.

Example, the weak law of large numbers

Proposition. Let $X_1, \dots, X_n$ be i.i.d. copies of a real X whose characteristic function is differentiable at 0 with $\phi'_X(0) = i\mu$, and let $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. Then $\bar{X}_n \xrightarrow{d} \mu$, and hence $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ by the theorem on constant limits in Part V.

Proof. By independence, property (xi) and the differentiability at 0 with $\phi_X(0) = 1$, we have

$$\phi_{\bar{X}_n}(t) = \phi_X\left(\frac{t}{n}\right)^n = \left(1 + \frac{it\mu}{n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n \rightarrow \infty} e^{it\mu} = \phi_{\delta_\mu}(t),$$

which is continuous, so that Levy item 2 concludes the proof.

Remarks on the law of large numbers

If $X \in L^1$, then ϕ_X is C^1 with $\phi'_X(0) = i\mathbb{E}_X[X]$ by property (vi), so that the proposition applies with $\mu = \mathbb{E}_X[X]$ and recovers the usual weak law.

Theorem (strong law of large numbers). Let $(X_i)_{i \in \mathbb{N}}$ be i.i.d. copies of a random vector X of $\mathbb{R}^d$ with $\mathbb{E}_X[\|X\|] < \infty$, and let $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. Then $\bar{X}_n \xrightarrow{a.s.} \mathbb{E}_X[X]$.

Proof omitted. This result is strictly stronger than the weak law, and its proof relies on a truncation and a Borel and Cantelli argument rather than on a Fourier argument, since almost sure convergence is not a property of the laws alone (see van der Vaart, Proposition 2.16).

Limits of the Fourier proof. It only manipulates $\mathbb{P}_{\bar{X}_n}$, so it can never produce an almost sure statement, which is the structural reason why the two laws of large numbers have unrelated proofs.

Example, the central limit theorem

Theorem. Let $X_1, \dots, X_n$ be i.i.d. copies of a real X with $\mathbb{E}_X[X] = 0$ and $\mathbb{E}_X[X^2] = 1$. Then $\sqrt{n}\bar{X}_n \xrightarrow{d} \mathcal{N}(0, 1)$.

Proof. By property (vi) with $r = 2$, ϕ_X is C^2 with $\phi_X(0) = 1$, $\phi'_X(0) = i\mathbb{E}_X[X] = 0$ and $\phi''_X(0) = -\mathbb{E}_X[X^2] = -1$, so that $\phi_X(u) = 1 - \frac{u^2}{2} + o(u^2)$ near 0. Hence,

$$\phi_{\sqrt{n}\bar{X}_n}(t) = \phi_X\left(\frac{t}{\sqrt{n}}\right)^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \xrightarrow{n \rightarrow \infty} e^{-t^2/2},$$

the characteristic function of $\mathcal{N}(0, 1)$, which is continuous, so that Levy item 2 concludes the proof.

Role of the hypotheses. Centering cancels the first order term, while unit variance fixes the scale.

The central limit theorem in $\mathbb{R}^d$

Theorem. Let $X_1, \dots, X_n$ be i.i.d. copies of a random vector X of $\mathbb{R}^d$ with $\mathbb{E}_X[\|X\|^2] < \infty$, mean μ and covariance $\Sigma = \text{Cov}_X(X)$. Then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Proof. Let $t \in \mathbb{R}^d$. Applying the scalar theorem to the i.i.d. real variables $\langle t, X_i \rangle$, which have mean $\langle t, \mu \rangle$ and variance $t^\top \Sigma t$, we obtain

$$\langle t, \sqrt{n}(\bar{X}_n - \mu) \rangle = \sqrt{n} \left(\frac{1}{n} \sum_{i \leq n} \langle t, X_i \rangle - \langle t, \mu \rangle \right) \xrightarrow{d} \mathcal{N}(0, t^\top \Sigma t) \stackrel{d}{=} \langle t, \mathcal{N}(0, \Sigma) \rangle.$$

Since every projection converges, the Cramer and Wold device concludes the proof.

Remark. If $t^\top \Sigma t = 0$ for some $t \neq 0$, the projected limit is δ_0 , and the limiting gaussian is degenerate. This is not a problem, since $\mathcal{N}(0, \Sigma)$ was defined through its characteristic function.

Part V. Relations between the modes of convergence

The chain of implications

Theorem. For random vectors on a common probability space,

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X, \quad X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{\mathbb{P}} X.$$

Proof of the first. Let us fix $\epsilon > 0$. For almost every ω , $\|X_n(\omega) - X(\omega)\| \rightarrow 0$, so that $\mathbf{1}\{\|X_n(\omega) - X(\omega)\| \geq \epsilon\} \rightarrow 0$. Since the indicators are bounded by 1, dominated convergence yields

$$\mathbb{P}(\|X_n - X\| \geq \epsilon) = \int_{\Omega} \mathbf{1}\{\|X_n - X\| \geq \epsilon\} d\mathbb{P} \rightarrow 0.$$

Proof of the third. By Markov's inequality applied to $\|X_n - X\|^p$, we have $\mathbb{P}(\|X_n - X\| \geq \epsilon) \leq \epsilon^{-p} \mathbb{E}_{X_n, X}[\|X_n - X\|^p] \rightarrow 0$.

From convergence in probability to convergence in law

Let f be bounded by M and L -Lipschitz, and let $\epsilon > 0$. Splitting on whether X_n is close to X , we obtain

$$\begin{aligned} & |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \\ & \leq \mathbb{E}_{X_n, X}[|f(X_n) - f(X)|] && \text{Jensen} \\ & \leq \mathbb{E}_{X_n, X}[L\epsilon \mathbf{1}\{\|X_n - X\| \leq \epsilon\} + 2M\mathbf{1}\{\|X_n - X\| \geq \epsilon\}] && \text{Lipschitz, then bounded} \\ & \leq L\epsilon + 2M\mathbb{P}(\|X_n - X\| \geq \epsilon) && \text{monotonicity} \end{aligned}$$

Since the second term tends to 0, we obtain $\limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| \leq L\epsilon$ for every $\epsilon > 0$, hence the limsup is 0, and portmanteau (iii) concludes.

Counterexamples to the converse implications

In probability but not almost surely. On $\Omega = [0, 1]$ with Lebesgue measure, we consider the typewriter sequence, defined for $n = 2^k + j$ with $0 \leq j < 2^k$ by $X_n = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k}]}$. Then $\mathbb{P}(X_n \neq 0) = 2^{-k} \rightarrow 0$, whereas for every ω the sequence $X_n(\omega)$ takes the value 1 infinitely often and the value 0 infinitely often.

Almost surely but not in L^1 . We take $X_n = n\mathbf{1}_{[0, 1/n]}$ on the same space. Then $X_n \rightarrow 0$ almost surely, whereas $\mathbb{E}_{X_n}[X_n] = 1$ for every n .

In law but not in probability. With $U \sim \text{Unif}([0, 1])$, $X_n = U$ and $X = 1 - U$, as in the exercise of Part II, we have $X_n \xrightarrow{d} X$, while $\|X_n - X\| = |2U - 1|$ does not tend to 0 in probability.

Convergence to a constant

Theorem. If $X_n \xrightarrow{d} c$ for a constant $c \in \mathbb{R}^d$ then $X_n \xrightarrow{\mathbb{P}} c$.

Proof. Let $\epsilon > 0$. Since the set $F = \{x : \|x - c\| \geq \epsilon\}$ is closed and does not contain c , portmanteau (vi) yields

$$\limsup_n \mathbb{P}(\|X_n - c\| \geq \epsilon) = \limsup_n \mathbb{P}(X_n \in F) \leq \mathbb{P}(c \in F) = 0.$$

Combined with the previous theorem, this result shows that the two modes coincide for a constant limit, i.e. $X_n \xrightarrow{\mathbb{P}} c \iff X_n \xrightarrow{d} c$. This makes the weak law of large numbers usable, and explains why Slutsky's hypothesis is stated with a constant.

Asymptotically equivalent sequences

Theorem. If $X_n \xrightarrow{d} X$ and $\|X_n - Y_n\| \xrightarrow{\mathbb{P}} 0$, then $Y_n \xrightarrow{d} X$.

Proof. Let f be bounded by M and L -Lipschitz, and $\epsilon > 0$. Then

$$|\mathbb{E}_{Y_n}[f(Y_n)] - \mathbb{E}_X[f(X)]| \leq \underbrace{\mathbb{E}_{X_n, Y_n}[|f(Y_n) - f(X_n)|]}_{\leq L\epsilon + 2M\mathbb{P}(\|X_n - Y_n\| \geq \epsilon)} + \underbrace{|\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]|}_{\rightarrow 0 \text{ by portmanteau (iii)}},$$

so that the limsup is at most $L\epsilon$ for every $\epsilon > 0$, and portmanteau (iii) concludes the proof.

Reading. A perturbation that is negligible in probability is invisible in distribution. We use this result throughout the course, since every asymptotic normality proof ends by writing the estimator as a sum that satisfies a central limit theorem, plus a remainder that is $o_{\mathbb{P}}(1)$.

Combining two convergences

Theorem. Let (X_n) , (Y_n) , X , Y be random vectors and c a constant vector.

1. **(Slutsky)** If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathbb{P}} c$ then $(X_n, Y_n) \xrightarrow{d} (X, c)$.
2. If $X_n \xrightarrow{\mathbb{P}} X$ and $Y_n \xrightarrow{\mathbb{P}} Y$ then $(X_n, Y_n) \xrightarrow{\mathbb{P}} (X, Y)$.
3. If $X_n \xrightarrow{a.s.} X$ and $Y_n \xrightarrow{a.s.} Y$ then $(X_n, Y_n) \xrightarrow{a.s.} (X, Y)$.

Proof of the combination theorem

Item 1. Since $Y_n \xrightarrow{\mathbb{P}} c$, we have $\|(X_n, Y_n) - (X_n, c)\| = \|Y_n - c\| \xrightarrow{\mathbb{P}} 0$, so that, by the theorem on asymptotically equivalent sequences, it suffices to show $(X_n, c) \xrightarrow{d} (X, c)$. For f bounded continuous on $\mathbb{R}^d \times \mathbb{R}^{d'}$, the section $f_c(x) = f(x, c)$ is bounded continuous, so that

$$\mathbb{E}_{X_n}[f(X_n, c)] = \mathbb{E}_{X_n}[f_c(X_n)] \rightarrow \mathbb{E}_X[f_c(X)] = \mathbb{E}_X[f(X, c)].$$

Item 2. With the equivalent norm $\|(x, y)\| = \|x\| + \|y\|$, a union bound yields

$$\mathbb{P}(\|(X_n, Y_n) - (X, Y)\| \geq \epsilon) \leq \mathbb{P}(\|X_n - X\| \geq \frac{\epsilon}{2}) + \mathbb{P}(\|Y_n - Y\| \geq \frac{\epsilon}{2}) \rightarrow 0.$$

Item 3. The intersection of two events of probability 1 has probability 1, and both coordinates converge on it.

Exercise, insufficiency of marginal convergence

Claim. $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ do not imply $(X_n, Y_n) \xrightarrow{d} (X, Y)$.

Solution. Let X, Y be independent $\mathcal{N}(0, 1)$, and let us set $X_n = X$ and $Y_n = -X$ for every n . Then $Y_n \sim \mathcal{N}(0, 1) \stackrel{d}{=} Y$, so that both marginal convergences hold. However, $(X_n, Y_n) = (X, -X)$ is supported on the line $\{y = -x\}$, whereas (X, Y) has a density on $\mathbb{R}^2$, so that the two laws differ and $(X_n, Y_n) \xrightarrow{d} (X, Y)$ fails.

Contrast with Slutsky. When the second limit is a constant, the limit law on the product is forced to be $\mathbb{P}_X \otimes \delta_c$ whatever the dependence, since a degenerate coordinate cannot be correlated with anything.

The continuous mapping theorem

Theorem. Let $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be measurable, let C be the set of points at which g is continuous, and let X satisfy $\mathbb{P}(X \in C) = 1$. Then

$$1. X_n \xrightarrow{a.s.} X \implies g(X_n) \xrightarrow{a.s.} g(X).$$

$$2. X_n \xrightarrow{\mathbb{P}} X \implies g(X_n) \xrightarrow{\mathbb{P}} g(X).$$

$$3. X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X).$$

Remark. If $X = c$ is a constant, the condition reduces to the continuity of g at the single point c , which is the version we use in practice, together with Slutsky.

Proof, the almost sure and the in probability cases

Item 1. On the event $\{X_n \rightarrow X\} \cap \{X \in C\}$, which has probability 1, the continuity of g at $X(\omega)$ yields $g(X_n(\omega)) \rightarrow g(X(\omega))$.

Item 2. Let $\epsilon, \delta > 0$ and

$$B_\delta = \{x \in \mathbb{R}^d : \exists y, \|x - y\| \leq \delta \text{ and } \|g(x) - g(y)\| \geq \epsilon\}.$$

Splitting on $\{\|X_n - X\| \leq \delta\}$ and its complement, we obtain

$$\mathbb{P}(\|g(X_n) - g(X)\| \geq \epsilon) \leq \mathbb{P}(X \in B_\delta) + \mathbb{P}(\|X_n - X\| \geq \delta).$$

The second term tends to 0. For the first, every $x \in C$ leaves B_δ once δ is small enough, so that $\mathbf{1}\{x \in B_\delta \cap C\} \rightarrow 0$ as $\delta \rightarrow 0$, and dominated convergence yields $\mathbb{P}(X \in B_\delta) = \mathbb{P}(X \in B_\delta \cap C) \rightarrow 0$. Letting $n \rightarrow \infty$ first, then $\delta \rightarrow 0$, concludes the proof.

Proof, the convergence in law case

Item 3. Let $F \subset \mathbb{R}^m$ be closed. Then $\{g(X_n) \in F\} = \{X_n \in g^{-1}(F)\}$, and

$$g^{-1}(F) \subset \overline{g^{-1}(F)} \subset g^{-1}(F) \cup C^c.$$

Indeed, for the second inclusion, let $x \in \overline{g^{-1}(F)}$ and $x_k \rightarrow x$ with $g(x_k) \in F$. If $x \in C$, then $g(x_k) \rightarrow g(x)$, so that $g(x) \in F$ since F is closed, and $x \in g^{-1}(F)$. Otherwise, $x \in C^c$.

Applying portmanteau (vi) to the closed set $\overline{g^{-1}(F)}$,

$$\limsup_n \mathbb{P}(g(X_n) \in F) \leq \limsup_n \mathbb{P}(X_n \in \overline{g^{-1}(F)}) \leq \mathbb{P}(X \in \overline{g^{-1}(F)}) \leq \mathbb{P}(X \in g^{-1}(F)) + \underbrace{\mathbb{P}(X \in C^c)}_{=0},$$

which is $\mathbb{P}(g(X) \in F)$, and portmanteau (vi) in the other direction yields $g(X_n) \xrightarrow{d} g(X)$.

Slutsky, the usable form

Corollary. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} c$ with c constant, then

1. $X_n + Y_n \xrightarrow{d} X + c$, for $X_n, Y_n, c \in \mathbb{R}^d$,
2. $Y_n X_n \xrightarrow{d} cX$, for $X_n \in \mathbb{R}^d$ and $Y_n, c \in \mathbb{R}$,
3. $Y_n^{-1} X_n \xrightarrow{d} c^{-1} X$, for $X_n \in \mathbb{R}^d$ and $Y_n, c \in \mathbb{R} \setminus \{0\}$.

Proof. Since a constant limit in distribution is a limit in probability, the combination theorem yields $(X_n, Y_n) \xrightarrow{d} (X, c)$. We then apply continuous mapping with $g(x, y) = x + y$, then $g(x, y) = yx$, and then $g(x, y) = x/y$. The first two are continuous everywhere, and the third one on $\{y \neq 0\}$, which carries the limit law since $c \neq 0$.

Uniform convergence of the distribution functions

Lemma (Polya). If $X_n \xrightarrow{d} X$ in $\mathbb{R}$ and F_X is continuous on $\mathbb{R}$, then the convergence is uniform:

$$\sup_{x \in \mathbb{R}} |F_{X_n}(x) - F_X(x)| \xrightarrow{n \rightarrow \infty} 0.$$

Proof. Let us fix an integer N . Since F_X is continuous with limits 0 and 1, the intermediate value theorem yields $x_1 < \dots < x_{N-1}$ with $F_X(x_i) = i/N$, and we set $x_0 = -\infty, x_N = +\infty$. For $x \in [x_{i-1}, x_i]$, the monotonicity of F_{X_n} and of F_X gives

$$F_{X_n}(x_{i-1}) - F_X(x_{i-1}) - \frac{1}{N} \leq F_{X_n}(x) - F_X(x) \leq F_{X_n}(x_i) - F_X(x_i) + \frac{1}{N},$$

where we used $F_X(x_i) - F_X(x_{i-1}) = 1/N$ on both sides.

Hence, $\sup_x |F_{X_n}(x) - F_X(x)| \leq \max_{i \leq N} |F_{X_n}(x_i) - F_X(x_i)| + \frac{1}{N}$, and since this maximum is finite, $\limsup_n \sup_x |F_{X_n} - F_X| \leq 1/N$ for every N .

Convergence of moments

Let $\mathbb{P}(X_n = n) = 1/n$ and $\mathbb{P}(X_n = 0) = 1 - 1/n$. Then $X_n \xrightarrow{d} 0$, while $\mathbb{E}_{X_n}[X_n] = 1$ for every n . Note that portmanteau only tests bounded functions, and the identity is not bounded.

Definition. A sequence (X_n) is uniformly integrable if

$$\lim_{M \rightarrow \infty} \sup_{n \in \mathbb{N}} \mathbb{E}_{X_n}[\|X_n\| \mathbf{1}\{\|X_n\| \geq M\}] = 0.$$

Theorem. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be continuous on a set C , let $X \in C$ almost surely, and let $X_n \xrightarrow{d} X$. If $(f(X_n))_n$ is uniformly integrable then $\mathbb{E}_{X_n}[f(X_n)] \rightarrow \mathbb{E}_X[f(X)]$.

Proof, with a criterion for uniform integrability

We assume $f \geq 0$, the general case following by splitting into positive and negative parts. For every $M > 0$,

$$\begin{aligned} \limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_X[f(X)]| &\leq \limsup_n |\mathbb{E}_{X_n}[f(X_n)] - \mathbb{E}_{X_n}[f(X_n) \wedge M]| \\ &\quad + \limsup_n |\mathbb{E}_{X_n}[f(X_n) \wedge M] - \mathbb{E}_X[f(X) \wedge M]| \\ &\quad + |\mathbb{E}_X[f(X) \wedge M] - \mathbb{E}_X[f(X)]|. \end{aligned}$$

The first line is at most $\sup_n \mathbb{E}_{X_n}[f(X_n)\mathbf{1}\{f(X_n) \geq M\}]$, which is small for M large by uniform integrability, and the third one is small for M large by monotone convergence. The middle one is 0 for every M , since $f \wedge M$ is bounded and continuous on C , so that continuous mapping and portmanteau (ii) apply.

Criterion. If $\sup_n \mathbb{E}_{X_n}[\|X_n\|^{1+\eta}] < \infty$ for some $\eta > 0$, then (X_n) is uniformly integrable, since $\mathbb{E}_{X_n}[\|X_n\|\mathbf{1}\{\|X_n\| \geq M\}] \leq M^{-\eta}\mathbb{E}_{X_n}[\|X_n\|^{1+\eta}]$. A uniformly bounded moment of order strictly above 1 is what we check in practice.

Part VI. A first example in statistics

A proportion

Let $X_1, \dots, X_n$ be i.i.d. $\text{Ber}(p)$ with $p \in (0, 1)$ unknown, and $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$. We ask what can be said about p from the sample, with no assumption beyond the model.

Law of large numbers. Since $\mathbb{E}_X[X] = p$ and $X \in L^1$, we have $\bar{X}_n \xrightarrow{\mathbb{P}} p$. In other words, the estimator is consistent, which says nothing about its error.

Central limit theorem. Since $\text{Var}_X(X) = p(1 - p) < \infty$, we have

$$\sqrt{n}(\bar{X}_n - p) \xrightarrow{d} \mathcal{N}(0, p(1 - p)).$$

Hence, the error is of order $n^{-1/2}$ and its limiting law is known. However, it is not yet usable, since the scale $p(1 - p)$ involves the unknown.

Studentisation

Exercise. Show that

$$\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{\bar{X}_n(1 - \bar{X}_n)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Solution. Since $\bar{X}_n \xrightarrow{\mathbb{P}} p$, continuous mapping with $u \mapsto u(1 - u)$, which is continuous on $\mathbb{R}$, yields

$$\bar{X}_n(1 - \bar{X}_n) \xrightarrow{\mathbb{P}} p(1 - p).$$

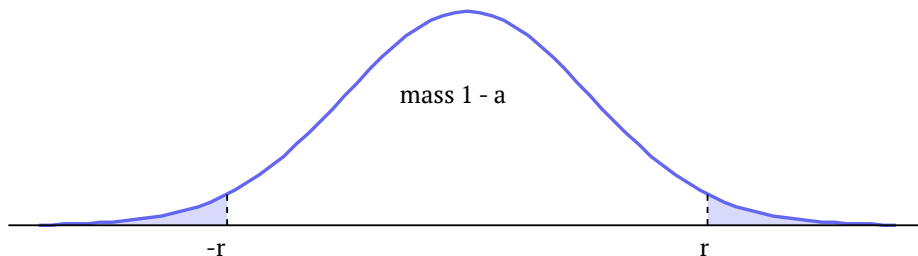
By Slutsky, we obtain the joint convergence

$$(\sqrt{n}(\bar{X}_n - p), \bar{X}_n(1 - \bar{X}_n)) \xrightarrow{d} (\mathcal{N}(0, p(1 - p)), p(1 - p)),$$

and continuous mapping with $g(u, v) = u/\sqrt{v}$ concludes, since g is continuous on $\{v > 0\}$, where the limit lies almost surely because $p \in (0, 1)$. The limit is thus $\mathcal{N}(0, p(1 - p))/\sqrt{p(1 - p)} \stackrel{d}{=} \mathcal{N}(0, 1)$.

An asymptotic confidence interval

Let r_α be the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, so that $\mathbb{P}(|Z| \leq r_\alpha) = 1 - \alpha$ for $Z \sim \mathcal{N}(0, 1)$.



By the previous slide and portmanteau (vii) applied to $B = [-r_\alpha, r_\alpha]$, whose boundary has zero mass under $\mathcal{N}(0, 1)$, we obtain

$$\mathbb{P} \left(\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{\bar{X}_n(1 - \bar{X}_n)}} \in [-r_\alpha, r_\alpha] \right) \xrightarrow{n \rightarrow \infty} 1 - \alpha.$$

Reading the interval

By inverting the event, we obtain that the random interval

$$I_n = \left[\bar{X}_n - r_\alpha \sqrt{\frac{\bar{X}_n(1 - \bar{X}_n)}{n}}, \bar{X}_n + r_\alpha \sqrt{\frac{\bar{X}_n(1 - \bar{X}_n)}{n}} \right] \text{ satisfies } \mathbb{P}(p \in I_n) \rightarrow 1 - \alpha.$$

It is computable. Every quantity in I_n is a function of the sample and of α , which is the purpose of the studentisation step, and this is why we use the plug-in estimate of the variance rather than $p(1 - p)$.

It is asymptotic. The statement is a limit rather than an inequality at fixed n . For a guarantee at fixed n , we need a concentration inequality, which leads to a worse constant and does not adapt to the variance.

It is not the only choice. Inverting instead $\{p : |\sqrt{n}(\bar{X}_n - p)| \leq r_\alpha \sqrt{p(1 - p)}\}$, a quadratic in p , leads to the Wilson interval, which is asymptotically equivalent and much better for small n or for p close to 0 or 1.

Part VII. The symbols $\mathcal{O}_{\mathbb{P}}$ and $\mathcal{O}_{\mathbb{P}}$

Two symbols

Definition. Let (X_n) be a sequence of random vectors and (R_n) a sequence of nonnegative random variables.

$X_n = o_{\mathbb{P}}(1)$ means $\|X_n\| \xrightarrow{\mathbb{P}} 0$. More generally, $X_n = o_{\mathbb{P}}(R_n)$ means that there is (Y_n) with $X_n = R_n Y_n$ for every n and $\|Y_n\| \xrightarrow{\mathbb{P}} 0$.

$X_n = O_{\mathbb{P}}(1)$ means (X_n) is uniformly tight. More generally, $X_n = O_{\mathbb{P}}(R_n)$ means that there is (Y_n) with $X_n = R_n Y_n$ for every n and (Y_n) uniformly tight.

The dictionary. The symbol $o_{\mathbb{P}}$ is the probabilistic version of "negligible", and $O_{\mathbb{P}}$ that of "bounded", with convergence in probability replacing convergence to 0 and uniform tightness replacing boundedness. Note that Prohorov item 1 states that $X_n \xrightarrow{d} X$ implies $X_n = O_{\mathbb{P}}(1)$, which is the source of almost every $O_{\mathbb{P}}$ in the course.

Transferring o and O to $o_{\mathbb{P}}$ and $O_{\mathbb{P}}$

Theorem. Let $X_n \xrightarrow{\mathbb{P}} 0$ in $\mathbb{R}^d$, let $q > 0$ and let $R : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be measurable with $R(0) = 0$. Then

1. $\|R(h)\| = o(\|h\|^q)$ as $h \rightarrow 0$ implies $R(X_n) = o_{\mathbb{P}}(\|X_n\|^q)$,
2. $\|R(h)\| = O(\|h\|^q)$ as $h \rightarrow 0$ implies $R(X_n) = O_{\mathbb{P}}(\|X_n\|^q)$.

Motivation. A Taylor expansion produces a deterministic remainder bound, valid as the increment goes to 0. However, every asymptotic expansion of an estimator evaluates that remainder at a random increment, and this theorem justifies the substitution.

Proof of the transfer

Let us define $g : \mathbb{R}^d \rightarrow \mathbb{R}^m$ by $g(h) = R(h)/\|h\|^q$ for $h \neq 0$ and $g(0) = 0$, so that $R(X_n) = g(X_n)\|X_n\|^q$ identically.

Item 1. The hypothesis $\|R(h)\| = o(\|h\|^q)$ means that $g(h) \rightarrow 0 = g(0)$ as $h \rightarrow 0$, so that g is continuous at 0. Since $X_n \xrightarrow{\mathbb{P}} 0$, continuous mapping yields $g(X_n) \xrightarrow{\mathbb{P}} 0$, which is the claim.

Item 2. By hypothesis, there are $M, \delta > 0$ with $\|R(h)\| \leq M\|h\|^q$ for $\|h\| \leq \delta$, hence $\|g(h)\| \leq M$ on that ball. Then

$$\limsup_n \mathbb{P}(\|g(X_n)\| \geq M) \leq \limsup_n \mathbb{P}(\|X_n\| \geq \delta) = 0,$$

so that $(g(X_n))$ is uniformly tight, and hence $R(X_n) = O_{\mathbb{P}}(\|X_n\|^q)$.

Exercise, $o_{\mathbb{P}}(O_{\mathbb{P}}(1)) = o_{\mathbb{P}}(1)$

Claim. If $X_n = O_{\mathbb{P}}(1)$ and $Y_n = o_{\mathbb{P}}(X_n)$, then $Y_n = o_{\mathbb{P}}(1)$.

Solution. By definition, there is (M_n) with $Y_n = M_n X_n$ and $M_n \xrightarrow{\mathbb{P}} 0$. Let us fix $\epsilon > 0$ and split on the size of X_n . For any $M > 0$, we have

$$\mathbb{P}(|Y_n| > \epsilon) = \mathbb{P}(|M_n X_n| > \epsilon, |X_n| \geq M) + \mathbb{P}(|M_n X_n| > \epsilon, |X_n| < M) \leq \mathbb{P}(|X_n| \geq M) + \mathbb{P}(|M_n| > \epsilon/M)$$

Given $\eta > 0$, we choose M with $\sup_n \mathbb{P}(|X_n| \geq M) \leq \eta$, which is possible because (X_n) is uniformly tight.

With M now fixed, the second term tends to 0 since $M_n \xrightarrow{\mathbb{P}} 0$. Hence, $\limsup_n \mathbb{P}(|Y_n| > \epsilon) \leq \eta$ for every $\eta > 0$.

The rules of calculation

The same two step split proves the rest of the calculus. For sequences of random vectors:

$o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1) + O_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$, by a union bound on the two halves of ϵ , respectively of M .

$o_{\mathbb{P}}(1) + O_{\mathbb{P}}(1) = O_{\mathbb{P}}(1)$ and $o_{\mathbb{P}}(1)O_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$, the latter being the exercise above.

$X_n \xrightarrow{d} X \implies X_n = O_{\mathbb{P}}(1)$, which is Prohorov item 1, and $X_n = O_{\mathbb{P}}(a_n)$ with $a_n \rightarrow 0$ implies $X_n = o_{\mathbb{P}}(1)$.

A warning. These are one way statements. In particular, nothing follows from $X_n = O_{\mathbb{P}}(1)$ about convergence of any kind, since tightness is a form of compactness rather than of convergence.